The Complete Structural Mereology of a Neurosymbolic Search-and-Rescue Manifold: A Unified Axiomatic System Specification

System Architecture Invariant Manifest

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1 Introduction and Architectural Core

We formalize the search-and-rescue plot for a central Fire-Station as an interlocking multi-graph Sequential Dynamic System (SDS) operating across distinct modal boundaries. Following Edward Zalta's Theory of Abstract Objects, the spatiotemporal execution plane maps to ordinary object exemplification (Fx), while the invariant control forests encode properties atemporally (xF). The ultimate objective of the system is the minimization of coordinate uncertainty to realize Mesarovic's Attainable Space (V'), culminating in physical vehicle extraction.

2 The 30-Dimensional Integrated Attribute Matrix

To inject continuous meaning and speech-act context into the message-passing convolutions, the GNN ingests an integrated, multi-scale node attribute matrix $\boldsymbol{X} \in \mathbb{R}^{N \times 30}$:

$$\boldsymbol{x}_i = \left[\boldsymbol{e}_i^{\text{emp (16)}} \mid \boldsymbol{g}_i^{\text{spatial (2)}} \mid \boldsymbol{p}_i^{\text{ppl (3)}} \mid \boldsymbol{r}_i^{\text{rsa (3)}} \mid \boldsymbol{s}_i^{\text{searle (3)}} \mid \boldsymbol{t}_i^{\text{deontic (3)}} \right] \in \mathbb{R}^{30} \tag{1}$$

3 Homomorphic Multi-Agent Lattices and Fitting Consensus

The architecture unifies the Epistemic Knowledge Lattice $\mathcal{L}_K$ and the Security Flow Lattice $\mathcal{L}_S$ into an identical branching partial-order structure governed by $a_k \in \mathcal{A}$. Parallel worker agents a_1 and a_2 occupy the same tier coordinate, formalizing a non-linear poset governed by the following strict axiomatic constraints:

$$a_3 \max\left(a_3 \preceq a_2\right), \quad a_3 \max\left(a_3 \preceq a_1\right), \quad a_2 \preceq a_0, \quad a_1 \preceq a_0 \tag{2}$$

4 Cross-Domain Isomorphic Mapping and Atemporal Control Planes

We formalize the mechanism for reasoning by analogy across disparate layers of reality by establishing structural isomorphisms between physical, socio-pragmatic, and scriptural situation spaces.

4.1 The Cross-Domain Isomorphism Operator

Let $\mathcal{D}_P$ represent the physical domain (SCADA grid power buses), $\mathcal{D}_S$ represent the socio-pragmatic domain (Searlean institutional structures), and $\mathcal{D}_R$ represent the scriptural-spiritual domain (salvific extraction trajectories). Following Barwise situation theory, while their extensional ground planes utilize distinct vocabularies and brute-fact tokens, their intensional relational forms are unified under an isomorphic mapping operator Φ:

$$\Phi : \mathcal{D}_P \cong \mathcal{D}_S \cong \mathcal{D}_R \tag{3}$$

This isomorphism guarantees that when the type-label stripping operator $\mathcal{S}_{\text{strip}}$ purges literal domain-specific tags, the underlying structural state vectors collapse to an identical geometric matrix configuration within the decision forest.

4.2 Atemporal Control and Semantic Validation

Let W_1 represent the spatiotemporal object execution plane where human actions and physical measurements reside, and let W_2 represent the Atemporal Controller encoding absolute, invariant control laws. Because any closed spatiotemporal system is structurally incomplete ($W_1 \nvdash v$), the system requires an external meta-language to achieve absolute semantic validation.

The Judith-Prodigal Son narrative serves as a canonical instantiation of this relationship. When stripped of extensional particulars, both narratives map to a shared non-convex trap topology $\mathcal{K}_{\text{trap}}$. The extraction trajectory is governed entirely by the properties encoded atemporally at the apex root node ($\top$) of the forest framework:

$$g(m, u, v) \equiv \exists y[R_{\text{system}}(m, u, y) \wedge E(m, u, y, v)] \tag{4}$$

where $m \in W_2$ represents the divine control input vector that intersects the lower syntax boundaries of W_1. Reasoning by analogy from the physical tracking limits of an electrical grid to the spiritual mechanics of scriptural typology allows the multi-agent framework to evaluate non-convex human dilemmas under invariant, zero-uncertainty safe control boundaries.